

Fig. 16: Current versus frequency plot

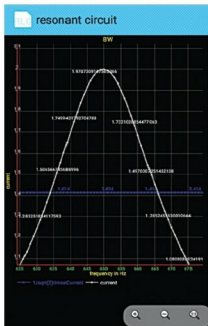


Fig. 17: Current versus frequency plot in Android after zoom in

EFY Note

The source code of this project is included in this month's EFY DVD and is also available for free download at source.efymag.com

quency, L and C act as a short circuit and only resistive element R is present. That is, $I_{max} = V/R = 8/4 = 2A$.

Bandwidth (BW). Bandwidth is based on current response. In Fig. 15, current gradually increases below resonance frequency and at resonance frequency, current is maximum and equal to resistor current, and above resonance frequency, current decreases. Below resonance frequency, frequency at which power is half its maximum value (current is $1/\sqrt{2}$ times of its maximum value), and this frequency is called lower cut-off frequency.

Above resonance frequency, frequency at which the power is half its maximum value (current is $1/\sqrt{2}$ times of its maximum value), and this frequency is called higher cut-off frequency. These two cut-off frequencies are called half-power frequencies.

$\text{Bandwidth} = \text{higher cut-off frequency} - \text{lower cut-off frequency}$ (BW = fH - fL).

In Figs 16 and 17, at resonance

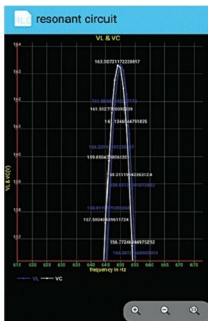


Fig. 18: VL and VC versus frequency plot

condition, current $I = 2A$, which is equal to resistor current $IR = 2A$ and lower cut-off frequency $f_L = 634.02\text{Hz}$ and the higher cut-off frequency $f_H = 665.85\text{Hz}$.

$\text{Bandwidth} = \text{higher cut-off frequency} - \text{lower cut-off frequency}$ (BW = $665.85\text{Hz} - 634.02 = 31.83\text{Hz}$).

Voltage at resonance frequency (VL and VC). In Fig. 18, at resonance frequency, voltage across L and C is

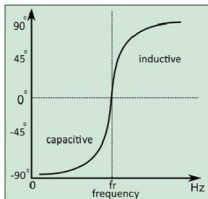


Fig. 19: Ideal case of angle versus frequency

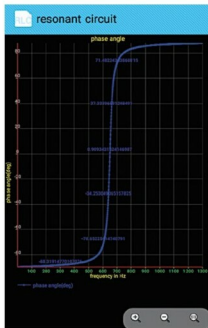


Fig. 20: Angle versus frequency plot

higher than the source voltage and VL and VC are in anti-phase and cancel each other out, resulting in a short circuit and supply current directly flowing through resistor R effectively.

Phase angle. The angle between voltage and current is called phase angle. In Figs 19 and 20, at resonance, the phase angle measures zero. The phase angle is positive for inductive and negative for capacitive.

In the concluding part, we will discuss the parallel RLC circuit.

To be continued next month



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